

Sathvik Ram Lanka

ramlanka1208@gmail.com | linkedin.com/in/sathvik-r-lanka | github.com/RamLanka05 | 630-935-1050

EDUCATION

University of Illinois Urbana-Champaign

December 2026

B.S. in **Computer Science and Statistics**, Minor in **Mathematics**

GPA: 3.81

- **Relevant Coursework:** Algorithms, Systems Programming, Database Systems, Data Structures, Languages & Automata, Machine Organization, Distributed Systems, Programming Languages & Compilers

EXPERIENCE

OLI Gardens, Patriotic Gardens

Aurora, IL

Software Engineer Intern

May 2025 – December 2025

- Shipped a full-stack MERN platform for a non-profit organization, deploying 15+ RESTful API endpoints using TypeScript, Node.js, and MongoDB, enabling secure content management and data orchestration across 3 core roles.
- Reduced unauthorized access attempts to zero by architecting a role-based access control system with JWT authentication and bcrypt password hashing, securing 50+ admin operations monthly.
- Increased donation processing capability by integrating Stripe payment gateways with custom tier logic and webhook handlers, enabling the organization to accept recurring and one-time contributions in production.
- Boosted frontend interactivity by developing custom React components with Tailwind CSS and Headless UI, including infinite-scroll carousels and filterable product sliders, improving user session duration by 35%.

ATLAS UIUC

Champaign, IL

Software Developer Intern

August 2025 – December 2025

- Shipped interactive Virtual Reality experiences for Quad Games, a Unity-based simulation of UIUC's Main Quad used for campus engagement, delivering optimized real-time graphics on Meta Quest 2 hardware.
- Reduced projectile simulation latency by 40% by implementing custom physics calculations in C# for a VR Archery Game, replacing naive collision detection with swept-sphere algorithms to handle edge cases at 90 FPS.
- Accelerated iteration cycles by configuring Meta Quest Link for rapid wireless testing and Visual Studio debugging workflows, reducing build-test-debug time from 8 minutes to 2 minutes per cycle.
- Solved critical stability issues in a physics-based VR demo for the PointVR research team by debugging edge-case floating-point precision errors, enabling seamless presentation to 50+ students in pilot testing.

PROJECTS

Rigveda Semantic Search Engine | Python, Hugging Face, Scikit-Learn, Pandas, Streamlit

- Architected an AI-native semantic search engine bridging ancient Sanskrit literature with modern NLP, enabling users to explore 10,000+ verses by conceptual meaning rather than rigid keyword matching.
- Engineered a robust data pipeline using Pandas and Parquet to transform nested JSON into a parallel corpus, leveraging Hugging Face embeddings and Scikit-Learn to deliver high-precision semantic retrieval via real-time Streamlit.
- Demonstrated deep understanding of vector embeddings and semantic similarity, implementing efficient retrieval mechanisms that rank results by cosine distance in high-dimensional vector space.

GPU-Accelerated Mandelbrot Set Generator | C++, CUDA, GPU Computation

- Achieved 150x speedup over CPU baseline by parallelizing the escape-time fractal algorithm across 2048 GPU threads in CUDA, rendering 4K UHD Mandelbrot images at interactive frame rates.
- Optimized I/O performance by replacing standard ASCII file streams with direct binary memory writing via `reinterpret_cast`, reducing 4K image serialization time from 12 seconds to under 1 second (92% improvement).

EarthDB: Full-Stack Geospatial Platform | React, TypeScript, Three.js, Python Flask, GCP, MySQL

- Engineered a full-stack 3D geospatial platform with real-time global data visualization in Three.js and a secure TypeScript backend, enabling authenticated CRUD operations for geospatial metadata.
- Architected a scalable GCP relational database aggregating 40,000+ geographic records, utilizing SQL triggers and stored procedures to optimize complex JSON queries and deliver live analytics dashboards.

TECHNICAL SKILLS

Languages: C/C++, Python, TypeScript, JavaScript, Go, SQL, C#, Java

Frameworks: React, Node.js, Express, Flask, CUDA, Three.js, Unity, Unreal Engine

Libraries & Databases: MongoDB, MySQL, TensorFlow, POSIX Threads, Hugging Face, Scikit-Learn

Software & Methodologies: Git, Docker, Google Cloud Platform, Databricks, Linux, GDB, AWS, Azure